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MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11732369 SELF TEST

This specification is approved for use by the US Army Armament Research and Dvelopment Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11732369 Self Test which is the A8 component of the M21 Ballistic Computer Unit, 11732400.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732369	Circuit Card Assembly: 11732369 Self Test

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750425	Adapter, Computer Board - A8
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert Plane/ Shock-Z
11821500	Vibration/Shock Test Fixture - Computer Unit
11821520	Automatic Test Assembly Central Processor Unit
11821854	Test Fixture Auto Card, Test Set, Computer, FAT

Packaging Data Sheet

P11732369	Packaging of Circuit Card Assembly: 11732369

(Copies of specitications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The self test circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732369 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification (see 4.5.1).

3.1.1 Signals. The assembly shall provide the following signals:

Signal Designation	Signal Nomenclature
a.	ST
b.	PWR OK
c.	RO (DC)
d.	R OK
e.	FB OK
f.	GCU OK
g.	RP OK
h.	OU OK
i.	RD OK and CU OK
j.	COMP FAIL
k.	RP FAIL and OU FAIL
l.	SYSTEM OK
m.	RF FAIL, GCU FAIL and CU FAIL

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	$73^{o} \pm 18^{o}F$
b.	Relative humidity	50 percent + 30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

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TABLE I. Loads, power and signal sources.

Item	Conditions	Characteristics	Connections
1.0	Loads		Connected between:
1.1	Resistor	510 ohm +5%, 10 watt (W) (or GE327 lamp or equivalent load)	P1-24A and 24 vdc (see item 2.1)
1.2	Resistor	510 ohm +5%, 10W (or GE327 lamp or equivalent load)	P1-20A and 24 vdc (see item 2.1)
1.3	Resistor	510 ohm +5%, 10W (or GE327 lamp or equivalent load)	P1-21B and 24 vdc (see item 2.1)
1.4	Resistor	510 ohm 5%, 10W (or GE327 lamp or equivalent load)	P1-20B and 24 vdc (see item 2.1)
1.5	Resistor	510 ohm 5%, 10W (or GE327 lamp or equivalent load)	P1-19A and 24 vdc (see item 2.1)
1.6	Resistor	510 ohm 5%, 10W (or GE327 lamp or equivalent load)	P1-23A and 24 vdc (see item 2.1)
1.7	Resistor	510 ohm 5%, 10W (or GE327 lamp or equivalent load)	P1-22A and 24 vdc (see item 2.1)
1.8	Resistor and Capacitor	100 ohm +5%, 1/4W, 2.2 microfrad (mfd) +10% in series	P1-12B (+) and P1-5A, -13A, -13B(-)
2.0	Power		Connected between:
2.1	24 vdc	Tolerance: +1.0 vdc Ripple $\leq$ 0.5 v p-p	P1-23B (+) and P1-19B (-); and items 1.1 through 1.7 to P1-23B (+)
2.2	15 vdc	Tolerance: +0.050 vdc Ripple $\leq$ 20 mv p-p	P1-6A (+) and P1-5A, -13A, -13B (-)

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TABLE I. Loads, power and signal sources. - Continued

Item	Conditions	Characteristics	Connections
2.3	10 vdc	Tolerance: +0.030 vdc Ripple ≤ 10 mv p-p	P1-3B (+) and P1-5A, -13A, -13B (-)
2.4	5 vdc	Tolerance: +0.5 vdc Ripple ≤ 0.1 v p-p	P1-1A (+) and P1-5A, -13A, -13B(-)
2.5	-10 vdc	Tolerance: +0.030 vdc Ripple ≤ 10 mv p-p	P1-25B (-) and P1-5A, -13A, -13B (+)
2.6	-15 vdc	Tolerance: +0.050 vdc Ripple ≤ 20 mv p-p	P1-21A (-) and P1-5A, -13A, -13B (+)
3.0	Signals		Connected between:
3.1	15 vdc (BIT)	Tolerance: +1.5 vdc Ripple ≤ 20 mv p-p	P1-15B and P1-5A, -13A, -13B (Rtn)
3.2	Variable: 0.00 to 6.50 vdc (PWR 1NH)	Tolerance: As specified Ripple ≤ 0.01 v p-p	P1-24B and P1-5A, -13A, -13B (Rtn)
3.3	15 to -15 v, 400 Hz square wave (MR1)	Tolerance: +1.5 v, +2 Hz Rise time (10% to 90%) < 1.0 microseconds (ms)	P1-7B and P1-5A, -13A -13B (Rtn)
3.4	15 to -15 v, 400 Hz square wave (MR2)	Tolerance: +1.5 v, +2 Hz Rise time (10% to 90%) < 1.0 ms Phase: 2	P1-7A and P1-5A, -13A, -13B (Rtn)
3.5	Variable: 0 to 5 vrms, 400 Hz (Ro(AC))	Tolerance: As specified and = 2 Hz Phase: 1	P1-2B and P1-5A, -13A, -13B (Rtn)

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TABLE I. Loads, power and signal sources. - Continued

Item	Conditions	Characteristics	Connections
3.6	1.000 vrms, 400 Hz (E feedback out)	Tolerance: ±0.010 vrms, ±2 Hz Phase: 1	P1-4B and P1-5A, -13A, -13B (Rtn)
3.7	Variable: 0 to 5 vrms, 400 Hz (D feedback out)	Tolerance: As specified and ±2 Hz Phase: 1	P1-4A and P-5A, -13A, -13B (Rtn)
3.8	Variable: 0 to 5 vrms, 400 Hz (GCU TEST)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-6B and P1-5A, -13A, -13B (Rtn)
3.9	Variable: 0 to 15 vrms, 400 Hz (D SERVO MOTOR 1)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-12A and P1-5A, -13A, -13B (Rtn)
3.10	Variable: 0 to 15 vrms, 400 Hz (D SERVO MOTOR 2)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-11B and P1-5A, -13A, -13B (Rtn)
3.11	Variable: 14 to +14 vdc (E MOTOR 1)	Tolerance: As specified	P1-16B and P1-5A, -13A, -13B (Rtn)
3.12	Variable: -14 to +14 vdc (E MOTOR 2)	Tolerance: As specified	P1-16A and P1-5A, -13A, -13B (Rtn)

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TABLE I. Loads, power and signal sources: - Continued

Item	Conditions	Characteristics	Connections
3.13	Variable: 0 to 5 vrms, 400 Hz (RX)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-10B and P1-5A, -13A, -13B (Rtn)
3.14	Variable: 0 to 5 vrms, 400 Hz (-epsilon)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-9A and P1-5A, -13A, -13B (Rtn)
3.15	Variable: 0 to 5 vrms, 400 Hz (CRS1)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-9B and P-15A, -13A, -13B (Rtn)
3.16	Variable: 0 to 5 vrms, 400 Hz (CRS2)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-8A and P1-5A, -13A, -13B (Rtn)
3.17	Variable: 0 to 5 vrms, 400 Hz (CRR1)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-14A and P1-5A, -13A, -13B (Rtn)
3.18	Variable: 0 to 5 vrms, 400 Hz (CRR2)	Tolerance: As specified and ±2 Hz Phase: As specified (1 or 2)	P1-14B and P1-5A, -13A, -13B (Rtn)
3.19	Variable: 0.5 or 4.5 vdc	Tolerance: +0.5 vdc Ripple $\leq$ 0.1 v p-p	P1-17B and P1-5A, -13A, -13B (Rtn)
3.20	Variable: 0.5 or 4.5 vdc	Tolerance: +0.5 vdc Ripple $\leq$ 0.1 v p-p	P1-18B and P1-5A, -13A, -13B (Rtn)
3.21	Variable: 0.0 or 4.5 vdc	Tolerance: 0.0 ±0.5 vdc 4.5±1.0 vdc Ripple $\leq$ 0.1 v p-p	P1-22B and P1-5A, -13A, -13B (Rtn)

NOTES:
1. Phase 1 signals in phase ±2.5° with item 3.3
2. Phase 2 signals in phase ±2.5° out-of-phase with item 3.3

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3.3.1 ST output. With the power (items 2.4 and 2.5) and the BIT signal (item 3.1) of table I applied, the output at P1-18A shall be 3.4 + 1.0 vdc. With the BIT signal removed from P1-15B, the output at P1-18A shall be 0.5 ± 0.5 vdc. (See 4.6.2.1)

3.3.2 PWR OK output. With the power (table I, item 2) applied, and:

a. When the table I, item 3.2, signal source is adjusted to between 0.00 and 3.40 vdc and applied, the output at P1-25A shall be 0.5 ± 0.5 vdc.

b. When the table I, item 3.2, signal source is adjusted to between 4.00 and 5.50 vdc and applied, the output at P1-25A shall be logic ONE (10.0 ± 1.0 vdc).

c. When the table I, item 3.2, signal source is adjusted to between 6.00 and 6.50 vdc and applied, the output at P1-25A shall be logic ZERO. (See 4.6.2.2)

3.3.3 Ro (DC) output. With the load (item 1.8), power (items 2.2 and 2.6) and MR2 signal (item 3.4) of table I applied, the nominal output at P1-12B (VDC) shall be described by the following transfer function.

$$V_{12B} = 0.90076 * V_{2B}$$

where:

a. V_{12B} is the output at P1-12B.

b. V_{2B} is the input at P1-2B (table I, item 3.5).

The tolerance of the output shall be not greater than the value of the following equation:

$$\Delta V_{12B} = 0.00105 * \left| V_{2B} \right| + 0.0085 \qquad \text{(See 4.6.2.3)}$$

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3.3.4 R OK output. With the power (items 2.2, 2.3, 2.4, and 2.6) and the MR2 signal (item 3.4 of table I) applied, and:

a. With the table I, item 3.5, signal source adjusted to 0.5 + 0.5 vrms and applied, the output at P1-3A shall be logic zero (0.5 ± 0.5 vdc).

b. When the table I, item 3.5, signal source is increased to 4.5 ± 0.5 vrms, the output at P1-3A shall switch to logic ONE (4.5 ± 1.0 vdc) and back to logic ZERO. The output shall switch to logic ONE at 2.671 ± 0.047 vrms and back to the logic ZERO at 2.765 ± 0.045 vrms.

c. When the table I, item 3.5, signal source is decreased to 0.5 ± 0.5 vrms, the output at P1-3A shall switch to logic ONE and back to logic ZERO. The output shall switch to logic ONE at 2.765 ± 0.045 vrms and back to logic ZERO at 2.671 ± 0.047 vrms. (See 4.6.2.4)

3.3.5 FB OK output. With the power (items 2.2 through 2.6) and MR1, MR2, and E feedback out signals (item 3.3, 3.4, and 3.6) of table I applied, and:

a. With the table I, Item 3.7, signal source adjusted to 0.5 + 0.5 vrms and applied, the output at P1-5B shall be logic ZERO (1.0 ± 0.5 vdc).

b. When the table I, item 3.7, signal source is increased to 4.5 ± 0.5 vrms, the output at P1-5B shall switch to logic ONE (4.5 ± 1.0 vdc) and back to logic ZERO. The output shall switch to logic ONE at 1.313 ± 0.025 vrms and back to logic ZERO at 1.586 ± 0.018 vrms.

c. When the table I, item 3.7, signal source is decreased to 0.5 ± 0.5 vrms, the output at P1-5B shall switch to logic ONE and back to logic ZERO. The output shall switch to logic ONE at 1.586 ± 0.018 vrms and back to logic ZERO at 1.313 ± 0.025 vrms. (See 4.6.2.5)

3.3.6 GCU OK output. With the power (items 2.2 through 2.6) and MR1 and MR2 signals (items 3.3 and 3.4) of table I applied, and:

a. With the table I, item 3.8, signal source adjusted to 4.5 ± 0.5 vrms, phase 2, and applied, the output at P1-8B shall be logic ZERO (0.5 - 0.5 vdc).

b. When the table I, item 3.8, signal source is increased to 4.5 ± 0.5 vrms, phase 1, the output at P1-8B shall switch to logic ONE (4.5 ± 1.0 vdc) and back to logic ZERO. The output shall switch to logic ONE at 0.066 ± 0.026 vrms, phase 2, and back to logic ZERO at 0.079 ± 0.023 vrms, phase 1.

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c. When the table I, item 3.8, signal source is decreased to 4.5 ± 0.5 vrms, phase 2, the output at P1-8B shall switch to logic ONE and back to logic ZERO. The output shall switch to logic ONE at 0.079 ± 0.023 vrms, phase 1, and back to logic ZERO at 0.066 ± 0.026 vrms, phase 2. (See 4.6.2.6)

3.3.7 RP OK output. With the power (items 2.2, 2.4, and 2.6) and MR1 and MR2 signals (items 3.3 and 3.4) of table I applied, and:

a. With the table I, item 3.10, signal source adjusted to 0.000 ± 0.010 vrms and applied, and with the table I, item 3.9, signal source adjusted to 14.5 ± 0.5 vrms, phase 2, and applied, the output at P1-15A shall be logic ZERO (0.5 ± 0.5 vdc).

b. When the table I, item 3.9, signal source is increased to 14.5 ± 0.5 vrms, phase 1, the output at P1-15A shall switch to logic ONE (4.5 ± 1.0 vdc) and back to logic ZERO. The output shall switch to logic ONE at 9.645 ± 0.897 vrms, phase 2, and back to logic ZERO at 9.919 ± 1.053 vrms, phase 1.

c. When the table I, item 3.9, signal source is decreased to 14.5 ± 0.5 vrms, phase 2, the output at P1-15A shall switch to logic ONE and back to logic ZERO. The output shall switch to logic ONE at 9.919 ± 1.053 vrms, phase 1, and back to logic ZERO at 9.645 ± 0.897 vrms, phase 2.

d. When the table I, item 3.9, signal source is increased to 0.000 ± 0.010 vrms, and the table I, item 3.10, signal source is decreased to 14.5 ± 0.5 vrms, phase 2, the output at P1-15A shall be logic ZERO.

e. When the table I, item 3.10 signal source is increased to 14.5 ± 0.5 vrms, phase 1, the output at P1-15A shall switch to logic ONE and back to logic ZERO. The output shall switch to logic ONE at 9.919 ± 1.053 vrms, phase 2, and back to logic ZERO at 9.645 ± 0.897 vrms, phase 1.

f. When the table I, item 3.10, signal source is decreased to 14.5 ± 0.5 vrms, phase 2, the output at P1-15A shall switch to logic ONE and back to logic ZERO. The output shall switch to logic ONE at 9.645 ± 0.897 vrms, phase 1, and back to logic ZERO at 9.919 ± 1.053 vrms, phase 2. (See 4.6.2.7)

3.3.8 OU OK output. With the power (items 2.2, 2.4, and 2.6) of table I applied, and:

a. With the table I, item 3.12, signal source adjusted to 0.000 ± 0.010 vdc and applied, and with the table I, item 3.11, signal source adjusted to -14.0 ± 0.5 vdc and applied, the output at P1-17A shall be logic ZERO (0.5 ± 0.5 vdc).

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b. When the table I, item 3.11, signal source is increased to 14.0 ± 0.5 vdc, the output at P1-17A shall switch to logic ONE (4.5 ± 05 vdc) and back in logic ZERO. The output shall switch to logic ONE at -11.790 ± 1.119 vdc and back to logic ZERO at 12.035 ± 1.157 vdc.

c. When the table I, item 3.11, signal source is decreased to -14.0 ± 0.5 vdc, the output at P1-17A shall switch to logic ONE and back to logic ZERO. The output shall switch to logic ONE at 12.035 ± 1.157 vdc and back to logic ZERO at -11.790 ± 1.119 vdc.

d. When the table I, item 3.11, signal source is adjusted to 0.000 ± 0.010 vdc, the output at P1-17A shall be logic ONE.

e. When the table I, item 3.12, signal source is increased to 14.0 ± 0.5 vdc, the output at P1-17A shall switch to logic ZERO. The output shall switch to logic ZERO at 11.790 ± 1.119 vdc.

f. When the table I, item 3.12, signal source is decreased to -14.0 ± 0.5 vdc, the output at P1-17A shall switch to logic ONE and back to logic ZERO. The output shall switch to logic ONE at 11.790 ± 1.119 vdc and back to logic ZERO at -12.035 ± 1.157 vdc.

g. When the table I, item 3.12, signal source is increased to 0.000 ± 0.010 vdc, the output at P1-17A shall switch to logic ONE. The output shall switch to logic ONE at -12.035 ± 1.157 vdc. (See 4.6.2.8)

3.3.9 RD OK and CU OK outputs. - With the power (items 2.2 through 2.6) and MR1 and MR2 signals (items 3.3 and 3.4) of table I applied, and:

a. With the table I, items 3.13 through 3.18, signal sources adjusted to 0.000 ± 0.010 vrms and applied, the outputs at P1-11A and P1-10A shall be logic ONE (4.5 ± 1.0 vdc).

b. When the table I, item 3.13, signal source is increased to 4.5 ± 0.5 vrms, phase 1, the output at P1-10A shall switch to logic ZERO (0.5 ± 0.5 vdc). The output shall switch to logic ZERO at 0.783 ± 0.112 vrms, phase 1.

c. When the table I, item 3.13, signal source is decreased to 4.5 ± 0.5 vrms, phase 2, the output at P1-10A shall switch to logic ONE and the output at P1-11A shall switch to logic ZERO. The output at P1-10A shall switch to logic ONE at 0.783 ± 0.112 vrms, phase 1, and the output at P1-11A shall switch to logic ZERO at 0.562 ± 0.066 vrms, phase 2.

d. When the table I, item 3.13, signal source is increased to 0.000 ± 0.010 vrms, the output at P1-11A shall switch to logic ONE. The output shall switch to logic ONE at 0.562 ± 0.066 vrms, phase 2. (See 4.6.2.9)

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3.3.10 RD and CU amplifier resistors. - With the power (items 2.2 through 2.6), MR1 and MR2 signals (items 3.3 and 3.4), and RD/CU amplifier resistor input signals (Items 3.13 through 3.18) of table I applied, and with P1-10A interconnected with P1-5A, the nominal output at Test Point Four (TP4, vdc) shall be described by the following transfer function:

$$V_{TP4} = -1.35045 * V_{10B} - 5.512 * V_{9A} - 5.4018 * V_{9B}$$
$$-5.4018 * V_{8A} - 2.7009 * V_{14A} - 2.7009 * V_{14B}$$
$$+0.043978 * V_{10A} + V_{TP4}(0)$$

where:

a. V_{TP4} is the output to TP4.

b. V_{10B} is the input at P1-10B (table I, item 3.13)

c. V_{9A} is the input at P1-9A (table I, item 3.14)

d. V_{9B} is the input at P1-9B (table I, item 3.15)

e. V_{8A} is the input at P1-8A (table I, item 3.16)

f. V_{14A} is the input at P1-14A (table I, item 3.17)

g. V_{14B} is the input at P1-14B (table I, item 3.18)

h. V_{10A} is the output at P1-10A

i. $V_{TP4}(0)$ = voltage measured at test point 4 (TP4, vdc) when all inputs are adjusted to 0.000 $\pm$ 0.010 vrms.

The tolerance of the output shall be not greater than the value of the following equation:

$$\Delta V_{TP4} = 0.01861 * \left|V_{TP4}\right| - + 0.00068 * \left|V_{10B}\right|$$
$$+0.00276 * \left|V_{9A}\right| + 0.00270 * \left|V_{9B}\right|$$
$$+0.00270 * \left|V_{8A}\right| + 0.00135 * \left|V_{14A}\right|$$
$$+0.00135 * \left|V_{14B}\right| + 0.00406 * \left|V_{10A}\right| + 0.010$$

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TABLE II. Comp fail output.

Item	Input Conditions at Pins			
	P1-1A 1/, 8/ (D Feedback Out)	P1-16B 2/, 8/ (E Motor 1)	P1-12A 3/, 8/ (D Servo Motor 1)	P1-17B 4/, 8/ (MDE OK)
1	4.5 ± 0.5 Vrms	14.0 ± 0.5 Vdc	14.5 ± 0.5 Vrms, Ø1	4.5 Vdc
2	4.5 ± 0.5 Vrms	14.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	4.5 Vdc
3	4.5 ± 0.5 Vrms	0.0 ± 0.5 Vdc	14.5 ± 0.5 Vrms, Ø1	4.5 Vdc
4	1.405 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	4.5 Vdc
5	1.405 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	4.5 Vdc
6	1.405 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.5 Vdc
7	N/A	N/A	N/A	0.5 Vdc
8	N/A	N/A	N/A	0.5 Vdc
9	N/A	N/A	N/A	N/A
10	1.465 ± 0.20 Vrms	14.0 ± 0.5 Vdc	14.5 ± 0.5 Vrms, Ø1	4.5 Vdc
11	1.465 ± 0.20 Vrms	14.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	4.5 Vdc
12	1.465 ± 0.020 Vrms	0.0 ± 0.5 Vdc	14.5 ± 0.5 Vrms, Ø1	4.5 Vdc

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TABLE II. Comp fail output. - Continued

Item	Input Conditions at Pins				Output Conditions at Pin P1-24A (Comp Fail)
	P1-10B 5/, 8/ ($\overline{Rx}$)	P1-15B 6/ (Bit)	P1-18B 7/ ($\overline{RF\ OK}$)	P1-22B 9/ (R.P. Pwr $\overline{Fail}$)	
1	0.000 ± 0.020 Vrms	15V	4.5 Vdc	0.0 Vdc	0.7 Vdc
2	0.000 ± 0.020 Vrms	15V	4.5 Vdc	0.0 Vdc	21 ± 2 Vdc
3	0.000 ± 0.020 Vrms	15V	4.5 Vdc	0.0 Vdc	24 ± 2 Vdc
4	0.000 ± 0.020 Vrms	15V	4.5 Vdc	0.0 Vdc	21 ± 2 Vdc
5	1.980 ± 0.020 Vrms, 02	15V	4.5 Vdc	0.0 Vdc	0.7 Vdc
6	0.000 ± 0.020 Vrms	15V	4.5 Vdc	0.0 Vdc	0.7 Vdc
7	N/A	Gnd or Open	4.5 Vdc	0.0 Vdc	21 ± 2 Vdc
8	N/A	15V	0.5 Vdc	0.0 Vdc	24 ± 2 Vdc
9	N/A	N/A	0.5 Vdc	4.5 Vdc	0.7 Vdc
10	0.000 ± 0.20 Vrms	15V	4.5 Vdc	0.0 Vdc	0.7 Vdc
11	0.000 ± 0.20 Vrms	15V	4.5 Vdc	0.0 Vdc	0.7 Vdc
12	0.000 ± 0.20 Vrms	15V	4.5 Vdc	0.0 Vdc	0.7 Vdc

1/ Table I, Item 3.7, signal.
2/ Table I, Item 3.11, signal.
3/ Table I, Item 3.9, signal.
4/ Table I, Item 3.19, signal.
5/ Table 1, Item 3.13, signal
6/ 15V - Interconnection with P1-6A, Open = No connection; Gnd = Interconnections with P1-5A.
7/ Table I, Item 3.20, signal.
8/ N/A = Indicates a not applicable condition.
9/ Table 1, Item 3.21, signal.

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Application of the transfer function and the tolerance equation shall be valid within the following limits:

a. $|V_{TP4}| < 0.1$ vdc

b. $|V_{10A}| < 0.1$ vdc

c. $|V_{10B}| < 5$ vrms

d. $|V_{9A}| < 5$ vrms

e. $|V_{9B}| < 5$ vrms

f. $|V_{8A}| < 5$ vrms

g. $|V_{14A}| < 5$ vrms

h. $|V_{14B}| < 5$ vrms

i. $|\text{OFFSET}| < 0.05$ vdc (See 4.6.2.10)

3.3.11 COMP FAIL output. With the load (item 1.1), power (items 2.1 through 2.6), and MR1, MR2, and E feedback out signals (items 3.3, 3.4, and 3.6) of table I applied, with the table I, item 3.12, signal source adjusted to 0.000 + 0.010 vdc and applied, and with the table I, items 3.10, and 3.14 through 3.18, signals sources adjusted to 0.000 + 0.020 vrms and applied, establishment in any item number order of the table II items input conditions shall result in the corresponding ordered output conditions specified. (See 4.6.2.11)

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TABLE III. RP FAIL and OU FAIL outputs.

Item	Input Conditions at Pins						Output Conditions at Pins	
	P1-4A 1/ (D Feedback Out)	P1-16B 2/ (E Motor 1)	P1-12A 3/ (D Servo Motor 1)	P1-22B 4/ (R.P. Pwr Fail)	P1-15B 5/ (Bit)	P1-18B 7/ (RF OK)	P1-20A 6/ (R.P. Fail)	P1-21B 6/ (O.U. Fail)
1	4.5 ± 0.5 Vrms	0.0 ± 0.5 Vdc	14.5 ± 0.5 Vrms, Ø1	0.0V	15V	4.5V	2Vdc	N/A
2	4.5 ± 0.5 Vrms	0.0 ± 0.5 Vdc	14.5 ± 0.5 Vrms, Ø1	0.0V	15V	0.5V	21 ± 2 Vdc	N/A
3	1.405 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.0V	15V	4.5V	24 ± 2 Vdc	24 ± 2 Vdc
4	4.5 ± 0.5 Vrms	0.0 ± 0.5 Vdc	14.5 ± 0.5 Vrms, Ø1	0.0V	Open or Gnd	4.5V	24 ± 2 Vdc	N/A
5	4.5 ± 0.5 Vrms	0.0 ± 0.5 Vdc	14.5 ± 0.5 Vrms, Ø1	4.5V	15V	4.5V	24 ± 2 Vdc	N/A
6	4.5 ± 0.5 Vrms	14.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.0V	15V	4.5V	N/A	2 Vdc
7	4.5 ± 0.5 Vrms	14.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.0V	15V	0.5V	N/A	24 ± 2 Vdc
8	1.405 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.0V	15V	4.5V	N/A	24 ± 2 Vdc
9	4.5 ± 0.5 Vrms	14.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	4.5V	15V	4.5V	N/A	24 ± 2 Vdc
10	4.5 ± 0.5 Vrms	14.0 ± 0.5 Vdc	0.0 ± 0.5 vrms	4.5V	Open or Gnd	4.5V	N/A	24 ± 2 Vdc

1/ Table I, Item 3.7 signal. 2/ Table I, Item 3.11, signal. 3/ Table I, Item 3.9, signal.
4/ Table I, Item 3.21, signal.
5/ 15V = Interconnection with P1-6A; Open= No connection; Gnd = Interconnections with P1-5A.
6/ N/A = Indicates a not applicable condition. 7/ Table I, Item 3.20, signal.

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3.3.12 RP FAIL and OU FAIL outputs. With the loads (items 1.2 and 1.3), power (items 2.1 through 2.6), and MR1, MR2, and E feedback out signals (items 3.3, 3.4, and 3.6) of table I applied, with the table I, item 3.12, signal source adjusted to 0.000 ± 0.010 vdc and applied, with the table I, items 3.10 and 3.13 through 3.18, signal sources adjusted to 0.000 ± 0.020 vrms and applied, and with the table I, item 3.19 signal source adjusted to 4.5 ± 1.0 vdc and applied, establishment in any item number order of the table III items input conditions shall result in the corresponding ordered output conditions specified. (See 4.6.2.12)

3.3.13 SYSTEM OK output. With the load (item 1.4), power (items 2.1 through 2.6), and MR1, MR2 and E feedback out signals (items 3.3, 3.4, and 3.6) of table I applied, with the table I, item 3.12, signal source adjusted to 0.000 ± 0.010 vdc and applied, with the table I, items 3.10 and 3.14 through 3.18, signal sources adjusted to 0.000 ± 0.020 vrms and applied, and with the table I, item 3.21, signal source adjusted to 0.0 vdc and applied, establishment in any item number order of the table IV items input conditions shall result in the corresponding ordered output conditions specified. (See 4.6.2.13)

3.3.14 RF FAIL and GCU FAIL and CU FAIL outputs. With the loads (items 1.5, 1.6, and 1.7), power (items 2.1 through 2.6) and MR1 and MR2 signals (items 3.3 and 3.4) of table I applied, and with the table I, items 3.14 through 3.18, signal sources adjusted to 0.000 ± 0.020 vrms and applied, establishment in any item number order of the table V items input conditions shall result in the corresponding ordered output conditions specified. (See 4.6.2.14)

3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 1. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

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TABLE IV. SYSTEM OK output.

Item	Input Conditions at Pins			
	P1-1A 1/, 7/ (D Feedback Out)	P1-16B 2/, 7/ (E Motor 1)	P1-12A 3/, 7/ (D Servo Motor 1)	P1-6B 4/ (G.C.U. Test)
1	1.465 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.000 ± 0.020 Vrms
2	1.465 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.000 ± 0.020 Vrms
3	1.465 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	4.5 ± 0.5 Vrms, Ø1
4	4.5 ± 0.5 Vrms	14.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.000 ± 0.020 Vrms
5	4.5 ± 0.5 Vrms	0.0 ± 0.5 Vdc,	14.5 ± 0.5 Vrms, Ø1	0.000 ± 0.020 Vrms
6	N/A	N/A	N/A	0.000 ± 0.020 Vrms
7	1.465 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.000 ± 0.020 Vrms
8	1.465 ± 0.020 Vrms	0.0 ± 0.5 Vdc	0.0 ± 0.5 Vrms	0.000 ± 0.020 Vrms

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TABLE IV. SYSTEM OK output. - Continued

Input Conditions at Pins					
Item	P1-15B 6/ (Bit)	P1-17B 6/, 7/ (MDE OK)	P1-18B 6/ (RF OK)	P1-16B 8/ (Rx)	Output Conditions at Pin P1-20B (System OK)
1	15V	5V	5V	0.000 ± 0.020 Vrms	2 Vdc
2	15V	5V	5V	1.980 ± 0.020 Vrms, 01	24 ± 2 Vdc
3	15V	5V	5V	0.000 ± 0.020 Vrms	24 ± 2 Vdc
4	15V	5V	5V	0.000 ± 0.020 Vrms	24 ± 2 Vdc
5	15V	5V	5V	0.000 ± 0.020 Vrms	24 ± 2 Vdc
6	15V	N/A	Gnd	0.000 ± 0.020 Vrms	24 ± 2 Vdc
7	15V	Gnd	5V	0.000 ± 0.020 Vrms	24 ± 2 Vdc
8	Open	5V	5V	0.000 ± 0.020 Vrms	24 ± 2 Vdc

1/ Table I, Item 3.7, signal.
2/ Table I, Item 3.11, signal.
3/ Table I, Item 3.9, signal.
4/ Table I, Item 3.8, signal.
5/ 15V = Interconnection with P1-6A; Open = No connection.
6/ 5V = Interconnection with P1-1A; Gnd = Interconnection with P1-5A.
7/ N/A = Indicates a not applicable condition.
8/ Table I, Item 3.13, signal.

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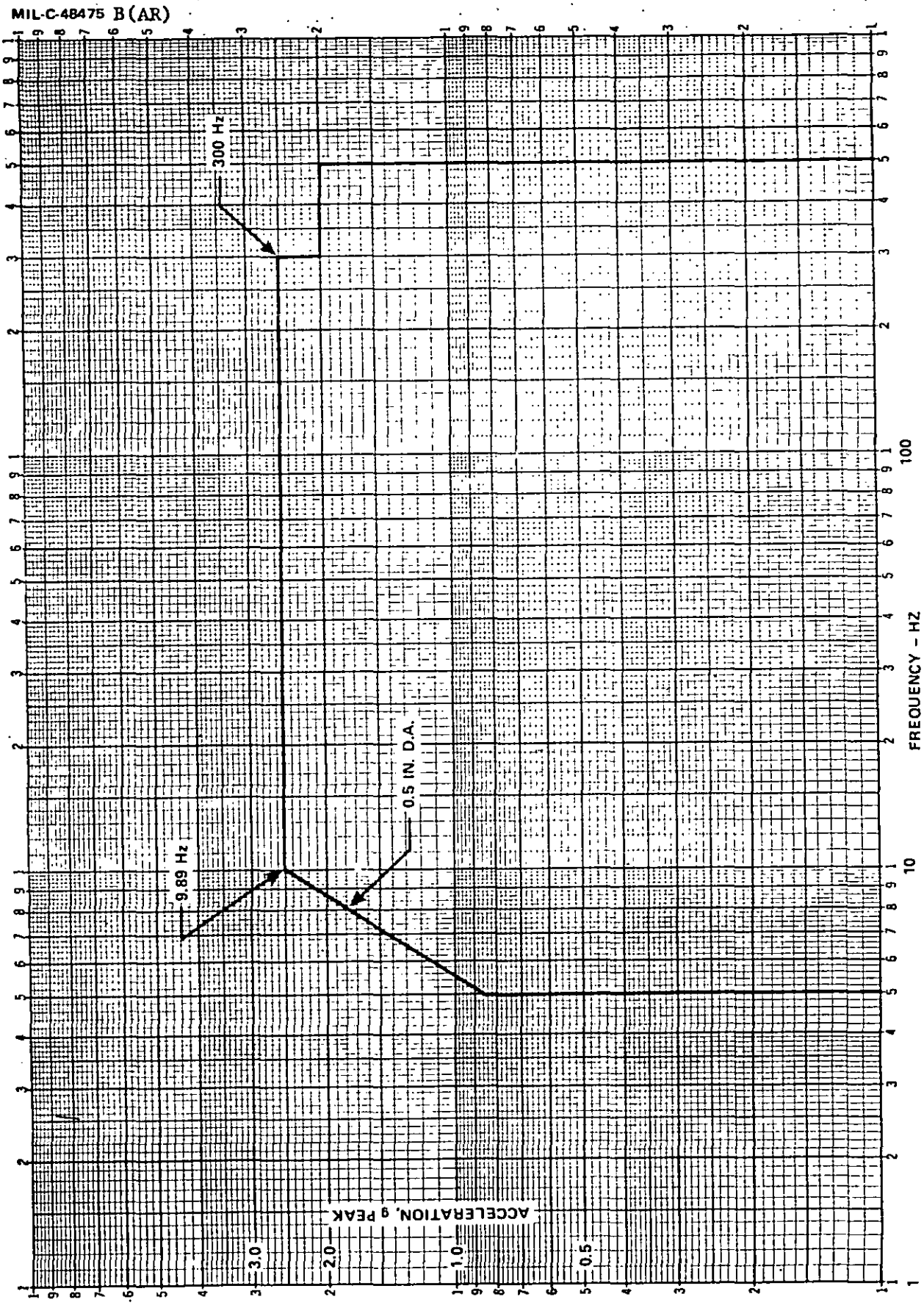


FIGURE 1. Vibration profile.

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TABLE V. RF and GCU and CU FAIL outputs.

Input Conditions at Pins				Output Conditions at Pin			
Item 6/	P1-18B 1/ (RF OK)	P1-6B 2/, 3/ (G.C.U. Test)	P1-10B 3/, 4/ (Rx)	P1-15B 5/ (Bit)	P1-19A 3/ (RF Fail)	P1-23A 3/ (GCU Fail)	P1-22A 3/ (CU Fail)
1	Gnd	4.5 ± 0.5 Vrms, 1	1.980 ± 0.020 Vrms, 01	15V	2 Vdc	24 ± 2 Vdc	24 ± 2 Vdc
2	Gnd	N/A	N/A	Open or Gnd	24 ± 2 Vdc	N/A	N/A
3	5V	4.5 ± 0.5 Vrms, 1	1.980 ± 0.020 Vrms, 01	15V	24 ± 2 Vdc	2 Vdc	2 Vdc
4	5V	4.5 ± 0.5 Vrms, 1	1.980 ± 0.020 Vrms, 01	Open or Gnd	N/A	24 ± 2 Vdc	24 ± 2 Vdc
5	5V	0.000 ± 0.020 Vrms	0.000 ± 0.020 Vrms	15V	N/A	24 ± 2 Vdc	24 ± 2 Vdc

1/ 5V = Interconnection with P1-1A; Gnd = Interconnection with P1-5A.
2/ Table I, Item 3.8, signal.
3/ N/A = Indicates a not application condition
4/ Table I, Item 3.13, signal.
5/ 15V = Interconnection with P1-6A; Open = No connection; Gnd = Interconnection with P1-5A.
6/ P1-17B = 5V (table I, item 2.4) for all tests.

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3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4, 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables VI, VII and VIII.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

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4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness the inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table VII.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table VIII shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table VII and only the test of item 307 in table VIII shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

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4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table VI and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unlesss otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732369.

TABLE VI. Performance tests.

Item	Characteristic	Requirement	Test Procedures
101	ST output	3.3.1	4.6.2.1
102	PWR OK output	3.3.2	4.6.2.2
103	Ro (DC) output	3.3.3	4.6.2.3
104	R OK output	3.3.4	4.6.2.4
105	FB OK output	3.3.5	4.6.2.5
106	GCU OK output	3.3.6	4.6.2.6
107	RP OK output	3.3.7	4.6.2.7
108	OU OK output	3.3.8	4.6.2.8
109	RD OK and CU OK outputs	3.3.9	4.6.2.9
110	RD and CU amplifier resistors	3.3.10	4.6.2.10

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TABLE VI. Performance tests. - Continued

Item	Characteristic	Requirement	Test Procedures
111	COMP FAIL outputs	3.3.11	4.6.2.11
112	RP FAIL and OU FAIL outputs	3.3.12	4.6.2.12
113	SYSTEM OK output	3.3.13	4.6.2.13
114	RF FAIL and GCU FAIL and CU FAIL outputs	3.3.14	4.6.2.14

TABLE VII. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

TABLE VIII. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature cycling	3.4.3 and 3.4.4	4.6.3.5

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4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table IX shall be used where specified to perform required tests.

4.6.1.1 AC voltage measurement. Unless otherwise specified, all ac voltage requirements shall be the in-phase component of the fundamental where MR1 (table I, item 3.3) is the phase reference.

TABLE IX. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750425	Adapter, Computer Board - A8
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
5	11821500	Vibration/Shock Test Fixture - Computer Unit
6	11821520	Automatic Test Assmebly Central Processor Unit
7	11821854	Test Fixture Auto Card, Test Set, Computer, FAT

4.6.2 Performance tests.

4.6.2.1 ST output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.1.

4.6.2.2 PWR OK output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.2.

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4.6.2.3 Ro (DC) output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.3.

4.6.2.4 R OK output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.4.

4.6.2.5 FB OK output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.5.

4.6.2.6 GCU OK output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.6.

4.6.2.7 RP OK output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.7.

4.6.2.8 OU OK output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.8.

4.6.2.9 RD OK and CU OK outputs. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.9.

4.6.2.10 RD and CU amplifier resistors. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.10.

4.6.2.11 COMP FAIL output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.11.

4.6.2.12 RP FAIL and OU FAIL outputs. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.12.

4.6.2.13 SYSTEM OK output. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.13.

4.6.2.14 RF FAIL and GCU FAIL and CU FAIL outputs. Use items 1, 2 and 6 of table IX or suitable equipment to verify the requirements of 3.3.14.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 5 of table IX or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 +4 g with a time duration of 18 +3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table VI.

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4.6.3.1.2 High intensity. Use items 3, 4 and 5 of table IX or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ±10.0 g with a time duration of 1.5 ±0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table VI.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Use items 4 and 5 of table IX or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 1 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table VI.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 7 of table IX or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table VI shall be applied.

4.6.3.4 Low temperature. Use item 7 of table IX or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table VI shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 2. At the conclusion of this test, subject the assembly to the performance tests of table VI.

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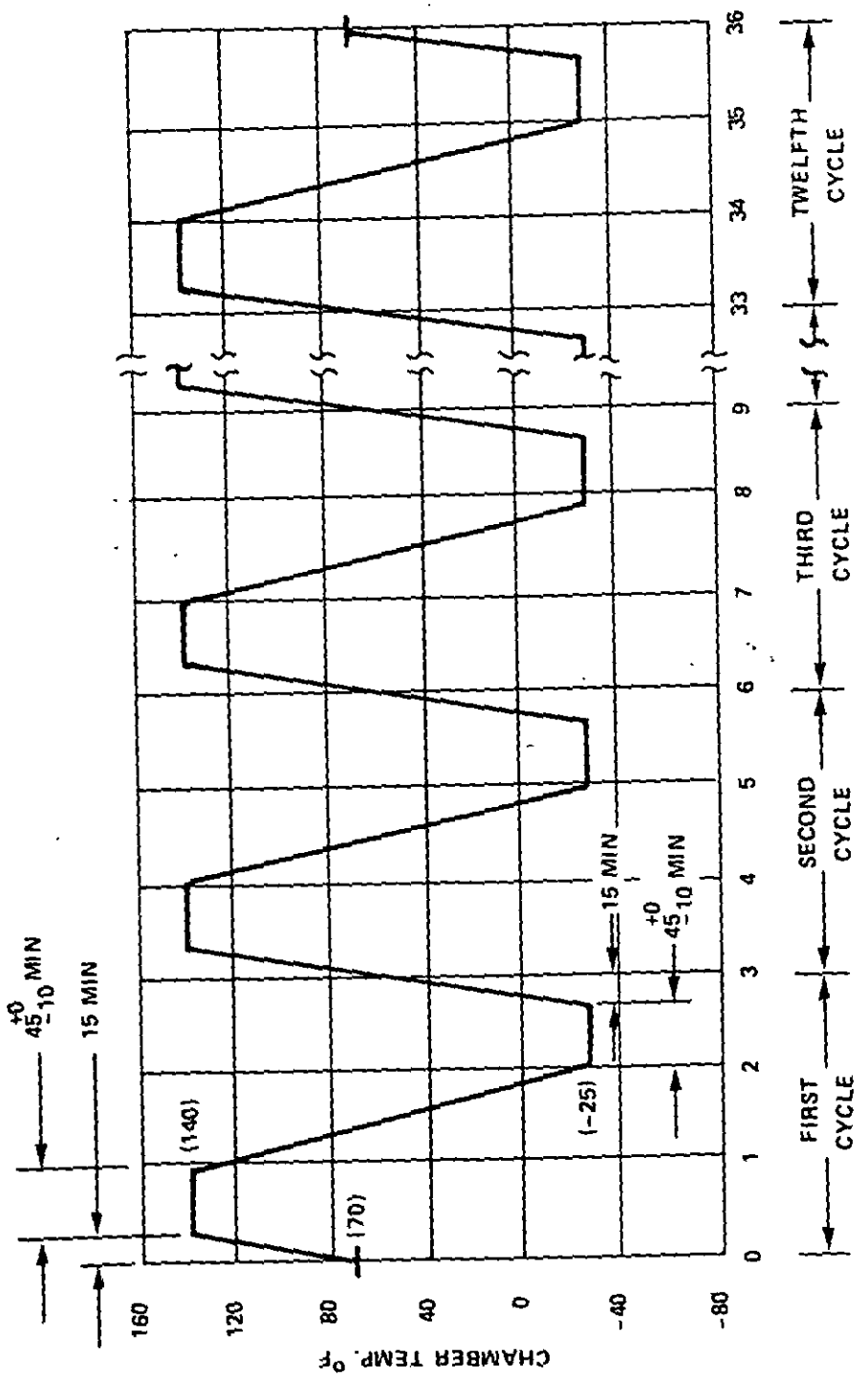


FIGURE 2. Temperature cycling.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732369. The level of protection shall be as specified in the procurement document. (see 4.5.6)

6. NOTES

6.1 Intended use. The self test circuit card assembly is used in the M21 Ballistic Computer Unit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Special tests and examination as required, the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48475B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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